

AI Peer Review Report

For the manuscript

Driving Sustainable Food Choices: The Impact of Specific and Actionable Product Swap Recommendations

Category	Score	Rating
Section Assessment	3.4/5	■■■
Rigor Assessment	3.6/5	■■■■
Writing Assessment	3.1/5	■■■
Final Score	3.4/5	■■■

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Executive Summary

This manuscript explores the effectiveness of providing specific, actionable product swap recommendations—primarily suggesting plant-based alternatives for animal-based products—in an online supermarket environment. The study further examines whether targeting the most impactful greenhouse gas emission (GHGE)-reducing swaps and offering GHGE reduction information enhances the intervention's effectiveness. Results show that most participants are receptive to swap suggestions, achieving an average 25% reduction in basket GHGE, with nearly 90% making at least one swap. However, neither targeted recommendations nor GHGE information significantly improved overall GHGE reduction for the majority, though GHGE information proved more influential among frequent meat consumers, highlighting the potential for tailored interventions in high-impact subgroups. A key strength of the manuscript lies in its timely focus on sustainable dietary behavior and its use of a controlled, ecologically valid online shopping experiment. The introduction provides a clear rationale for the intervention, referencing relevant literature on the limitations of generic dietary guidance. However, the manuscript's contribution is not explicitly articulated in relation to existing research, and the background literature review lacks depth in theoretical frameworks and critical synthesis. The introduction could better highlight the study's novelty and situate it within the broader context of digital behavioral interventions. Additionally, while the methodology is robust, the sample's limited diversity and the artificial setting constrain generalizability, and the statistical analysis would benefit from more rigorous assumption checks and corrections for multiple testing. To strengthen the manuscript, the authors should explicitly state the research gap and unique contribution in both the abstract and introduction, clarifying how this work advances the field. The background section should be expanded to critically engage with recent behavioral intervention studies, especially those employing digital or personalized strategies, and integrate relevant theoretical models. Methodologically, improving statistical rigor, clarifying terminology, and enhancing the reporting of model diagnostics are recommended. Finally, the discussion should more critically reflect on limitations, align conclusions with the main findings, and offer actionable implications for researchers, practitioners, and policymakers interested in promoting sustainable food choices.

Category	Sub-Category	Score
Section Assessment	S1 - Title and Keywords	4/5
	S2 - Abstract	3/5
	S3 - Introduction	3/5
	S4 - Literature Review	3/5
	S5 - Methodology	4/5
	S6 - Results	3/5
	S7 - Discussion	3/5
	S8 - Conclusion	3/5
	S9 - References	4/5
	S10 - Supplementary Materials	4/5
Rigor Assessment	R1 - Originality and Contribution	4/5
	R2 - Impact and Significance	4/5
	R3 - Ethics and Compliance	4/5
	R4 - Data and Code Availability	4/5
	R5 - Statistical Rigor	3/5
	R6 - Technical Accuracy	3/5
	R7 - Consistency	3/5
Writing Assessment	W1 - Language and Style	4/5
	W2 - Narrative and Structure	3/5
	W3 - Clarity and Conciseness	3/5
	W4 - Terminology Consistency	3/5
	W5 - Inclusive Language	3/5
	W6 - Citation Formatting	3/5
	W7 - Target Audience Alignment	3/5



S1 - Title and Keywords

Score: 4/5

The title effectively communicates the manuscript's focus on sustainable food choices and product swap interventions, but it is somewhat lengthy and lacks key outcome and context terms. The absence of a dedicated keywords section is a minor drawback. Improving the title's conciseness, adding relevant keywords (e.g., 'climate change', 'GHGE reduction', 'online experiment'), and optimizing for search engines would enhance discoverability and clarity. The current title adheres to academic standards but could be more impactful with better keyword integration and specificity.

Remarks	Original	Improved	Explanation
The title is somewhat lengthy and lacks explicit mention of key outcomes and the experimental context, reducing clarity and discoverability.	Driving Sustainable Food Choices: The Impact of Specific and Actionable Product Swap Recommendations	Enhancing Climate-Friendly Food Choices Through Targeted Product Swaps and GHGE Reduction: An Online Experiment	This revision consolidates the main focus, outcome, and experimental context into a concise, keyword-rich phrase, improving clarity, search engine optimization, and immediate relevance.
The title does not include high-impact keywords such as 'climate change', 'food choices', or 'behavior change', which are important for SEO and audience targeting.	Driving Sustainable Food Choices: The Impact of Specific and Actionable Product Swap Recommendations	Driving Climate-Friendly Food Choices: An Experimental Study of Targeted Product Swaps and Greenhouse Gas Emissions Reduction	Adding keywords like 'climate-friendly', 'experimental study', and 'GHGE reduction' improves discoverability and aligns the title with the manuscript's core contributions.
No dedicated keywords section is present, which is a missed opportunity for indexing and searchability.	No keywords section found	Include a keywords section with terms such as: 'sustainable diets', 'product swaps', 'greenhouse gas emissions', 'behavior change', 'online experiment', 'climate change', 'food choices'.	A keywords section enhances searchability and ensures the manuscript is indexed appropriately for relevant audiences.

S2 - Abstract

Score: 3/5

The abstract covers all major components—background, methods, results, and conclusions—but is overly detailed and dense, which affects readability and quick comprehension. Key hypotheses and outcomes are not clearly stated, and the language is technical, limiting accessibility for a broader audience. The implications of the findings are understated, and the structure could be improved by organizing content into clear sections. Simplifying language, clarifying main findings, and highlighting broader significance would improve the abstract's effectiveness.

Remarks	Original	Improved	Explanation
The abstract is overly lengthy and contains detailed methodological/statistical information that could be condensed for clarity.	This study examines the effectiveness of specific and actionable product swap recommendations - proposing plant-based alternatives in place of animal-based products - within an experimental online supermarket setting.	This study evaluates how specific, actionable product swap recommendations—such as replacing animal-based with plant-based products—affect consumers' climate-friendly food choices in an online supermarket experiment.	Rephrasing for clarity and brevity enhances readability and emphasizes the core research focus.
The abstract lacks explicit mention of key hypotheses and the specific primary and secondary outcomes.	Participants were overall receptive to receiving product swap recommendations. Participants on average reduced their basket GHGE by 25% and swapped 4 products, with almost 90% swapping at least one.	Participants generally responded positively to the recommendations, achieving an average reduction of 25% in basket greenhouse gas emissions (GHGE) and swapping an average of 4 products, with nearly 90% swapping at least one item.	Improves clarity by reducing redundancy and clarifying statistical measures for better readability.
The implications and significance of findings are somewhat understated; the abstract does not strongly communicate the potential real-world impact or policy relevance.	Our findings suggest that offering consumers the option to swap products—without the need for targeted recommendations or GHGE reduction information—may already be sufficient to encourage more sustainable purchasing behaviours for many.	Our results indicate that simply providing consumers with the option to swap products—without targeted suggestions or environmental information—may be enough to promote more sustainable purchasing behaviors for many individuals.	Enhances clarity and flow, making the statement more direct and accessible.

S3 - Introduction

Score: 3/5

The introduction provides a solid foundation with relevant literature and a clear focus on the environmental and health impacts of dietary choices. However, it lacks a concise statement of the specific research gap and does not clearly articulate the study's objectives or hypotheses upfront. The flow from background to objectives is somewhat disjointed, and the significance of the study could be better emphasized in relation to theory and policy. Tighter integration of background, explicit research questions, and improved structure would enhance clarity and impact.

Remarks	Original	Improved	Explanation
The research gap regarding consumer behavior interventions in real-world settings is not clearly articulated.	While broad dietary guidelines advocate for reducing animal-based food consumption, these recommendations often lack the specific and actionable direction to motivate behaviour change.	Although dietary guidelines recommend reducing animal-based foods, they frequently lack specific, actionable strategies that effectively motivate consumers to change their behaviour.	Clarifies the gap between general guidelines and the need for specific, actionable interventions, sharpening the research problem.
Objectives are embedded within the study design rather than clearly stated as research questions.	This study examines the effectiveness of specific and actionable product swap recommendations - proposing plant-based alternatives in place of animal-based products - within an experimental online supermarket setting.	This study investigates whether specific, actionable product swap recommendations—such as substituting animal-based products with plant-based alternatives—can effectively influence consumer choices in a realistic online shopping environment.	Adds clarity on the intervention and emphasizes the real-world setting, aligning with the study's novelty.
The flow from background to research aims is somewhat disjointed and could be improved for better readability.	The introduction could benefit from a more concise and integrated background that directly links environmental impacts, consumer behaviour challenges, and the specific research gap.	Streamline the background by briefly outlining the environmental and health impacts of food choices, then emphasizing the challenge of translating broad dietary guidelines into specific, actionable consumer interventions. Clearly state the existing gap in empirical evidence on real-world adoption of targeted product swaps, motivating the current study.	Enhances coherence and focus, making the background more relevant to the research aims.

S4 - Literature Review

Score: 3/5

The literature review covers recent empirical and simulation studies relevant to product swaps and GHGE reduction but lacks depth in theoretical frameworks, historical context, and coverage of broader environmental impacts. Critical engagement with conflicting evidence and discussion of mechanisms underlying consumer behavior change are limited. Organization and citation diversity could be improved. Expanding theoretical integration, updating sources, and providing critical synthesis would strengthen the review's comprehensiveness and analytical rigor.

Remarks	Original	Improved	Explanation
The review lacks comprehensive coverage of broader theoretical frameworks and historical development.	The study examines the effectiveness of specific and actionable product swap recommendations - proposing plant-based alternatives in place of animal-based products - within an experimental online supermarket setting.	Expand the literature review to include theoretical frameworks such as the Theory of Planned Behavior, Social Practice Theory, or Nudge Theory, which underpin consumer decision-making and behaviour change in sustainable food choices.	Incorporating these frameworks will provide a stronger conceptual foundation, situating the empirical findings within established theories and enhancing interpretative depth.
The review presents descriptive summaries but lacks critical engagement with conflicting evidence or mechanisms.	The review presents descriptive summaries of findings but lacks critical engagement with conflicting evidence, limitations of prior studies, or nuanced discussion of mechanisms underlying consumer behaviour change.	Incorporate critical appraisal of key studies, discussing methodological strengths and limitations, conflicting results, and gaps in understanding mechanisms of behaviour change.	This will improve analytical depth and demonstrate scholarly rigor, guiding future research directions.
References are somewhat outdated and lack diversity, missing recent reviews or meta-analyses.	While the review cites numerous relevant studies, many references are somewhat outdated or lack diversity in sources.	Update citations to include recent systematic reviews, meta-analyses, and key policy documents from the last 3-5 years, and diversify sources to include grey literature or reports from reputable organizations.	This will ensure the review reflects the latest evidence, enhancing credibility and relevance.

S5 - Methodology

Score: 4/5

The methodology is well-structured, with clear experimental design, randomization, and appropriate use of validated measures. However, the use of a simulated online supermarket limits ecological validity, and the sample is not fully representative, with over-representation of young males and exclusion of certain dietary groups. Statistical analyses are appropriate but lack robust correction for multiple testing and do not fully address potential confounders. Ethical procedures are mentioned but not detailed. Addressing these limitations would strengthen the study's validity and generalizability.

Remarks	Original	Improved	Explanation
The simulated environment limits ecological validity and generalizability.	The study employs a simulated online supermarket environment, which may not fully capture real-world shopping behaviors, especially over longer-term adoption and habitual change.	Incorporate field experiments or longitudinal studies in actual retail settings to validate whether the observed effects translate to real-world, habitual shopping behaviors over time.	This enhances ecological validity and ensures that findings are applicable beyond the experimental context.
The sample is restricted and not fully representative, limiting external validity.	The sample is restricted to German-speaking, non-vegan/non-vegetarian participants, with an over-representation of young males, which constrains the diversity and representativeness of the sample.	Expand recruitment to include more diverse populations across language, dietary preferences, age groups, and gender to improve representativeness and external validity.	A more diverse sample would allow for broader generalizations and better understanding of intervention effects across different demographic groups.
Statistical analysis lacks correction for multiple testing and consideration of additional confounders.	The primary outcome analysis relies heavily on regression models without robust correction for multiple testing or consideration of potential confounders beyond basic covariates.	Apply correction methods for multiple comparisons (e.g., Bonferroni or False Discovery Rate) and include additional relevant confounders such as baseline dietary habits or environmental attitudes to strengthen the validity of statistical inferences.	This reduces the risk of false positives and enhances the rigor and reliability of the analysis results.

S6 - Results

Score: 3/5

The results section presents appropriate analyses and relevant findings, but data presentation is dense and lacks sufficient explanation of effect sizes and practical significance. Model diagnostics and assumptions are not discussed, and interpretation of non-significant results is superficial. Tables and figures could be clarified with better legends and explanatory notes. Improvements in clarity, reporting of statistical assumptions, and deeper interpretation would enhance accessibility and scientific rigor.

Remarks	Original	Improved	Explanation
Presentation of regression results is cluttered and lacks explanation of effect sizes and practical relevance.	The regression model (Table 2 and Figure 2) indicated a significant intercept of 23% ($p < 0.001$), suggesting that participants in the reference condition already exhibited a substantial reduction in their basket GHGE.	Add a brief explanation of the intercept's meaning, e.g., 'The intercept of 23% indicates the estimated average GHGE reduction in the control group, reflecting baseline climate-friendly shopping behavior.'	Clarifies the interpretation of the regression output for readers unfamiliar with statistical modeling.
Models lack description of diagnostics or checks for statistical assumptions.	The models include multiple covariates but lack a description of diagnostics or assumptions.	Include a brief statement on model diagnostics performed (e.g., checks for multicollinearity, residual normality, heteroscedasticity) and their results.	Ensures transparency and confirms the robustness of the regression analyses.
Tables contain dense statistical data without explanations of key terms.	Tables contain dense statistical data without explanations of key terms.	Add footnotes or brief descriptions explaining key coefficients, such as what an odds ratio or IRR indicates in context.	Enhances accessibility for readers less familiar with statistical terminology, improving overall clarity.

S7 - Discussion

Score: 3/5

The discussion interprets the main findings and acknowledges limitations, but the analysis of non-significant results is superficial, and the comparison with existing empirical studies is limited. Reflection on biases from the experimental design and practical implementation challenges is insufficient. The section is somewhat repetitive and could benefit from clearer structure. Deeper analysis, explicit links to literature, and a more critical reflection on limitations and real-world application would strengthen the discussion.

Remarks	Original	Improved	Explanation
Interpretation of non-significant findings is superficial and lacks exploration of alternative explanations.	The regression analysis did not reveal any statistically significant effects of the Targeted or Information treatments on GHGE reduction compared to the reference group.	While the targeted and informational interventions did not significantly reduce overall basket GHGE, the increase in the selection of the most effective swap suggests some targeted influence on specific behaviors, which warrants further exploration.	Provides a more nuanced interpretation, acknowledging partial effects and avoiding overly simplistic conclusions.
The discussion lacks thorough comparison with existing empirical field or lab studies.	Our results indicate that offering specific and actionable product swap recommendations, derived from the final contents of baskets and presented before checkout, is an effective strategy.	Our findings support that providing consumers with specific, actionable product swap recommendations at the point of decision-making can effectively promote more climate-friendly purchasing behaviors, aligning with prior consumer preference research.	Enhances the connection to existing literature and emphasizes the practical relevance, improving contextual framing.
Reflection on limitations does not sufficiently analyze biases from the online, incentivized experimental design.	Despite these strengths, certain limitations must be acknowledged.	Despite the strengths of our experimental design, several limitations—including sample representativeness, ecological validity, and the focus solely on GHGE—should be considered when interpreting the findings.	Offers a more comprehensive and critical view, improving transparency and scientific rigor.

S8 - Conclusion

Score: 3/5

The conclusion summarizes key findings but does not fully clarify the strength of baseline and subgroup effects, nor does it explicitly state the degree to which research objectives were achieved. Practical implications are discussed but somewhat overstated without sufficient attention to limitations. The final statements are verbose and could be more concise. Enhancing clarity, balancing claims with caveats, and explicitly linking results to objectives would improve the conclusion's quality.

Remarks	Original	Improved	Explanation
The conclusion does not sufficiently clarify the strength of baseline and subgroup effects.	Participants demonstrated a general willingness to engage with product swap recommendations. On average, post-swap baskets exhibited a 24.9% (SD = 15.7) reduction in GHGE compared to pre-swap baskets.	Participants showed a strong overall engagement, with an average of 24.9% reduction in basket GHGE post-swap, indicating the intervention's potential effectiveness.	Clarifies the significance of the result and emphasizes the intervention's impact, strengthening the support statement.
The conclusion overstates the scalability and policy relevance without sufficiently addressing limitations.	The straightforward nature of this intervention, particularly when implemented without targeted strategies or GHGE information, makes it highly feasible for retailers operating online platforms or self-checkout systems.	This simple, scalable intervention can be easily integrated into online retail platforms or self-checkout systems, facilitating widespread adoption.	Strengthens the practical implications by emphasizing feasibility and scalability, aligning with policy relevance.
The final statement is somewhat verbose and repeats prior points, reducing clarity and impact.	In summary, providing consumers with specific and actionable product swap recommendations based on basket contents is an effective strategy for promoting climate-friendly food choices.	In summary, basket-specific, actionable product swap recommendations are a promising approach to promote sustainable food choices, especially when combined with targeted information for high-impact consumers.	Adds nuance and acknowledges subgroup differences, improving clarity and contribution understanding.

S9 - References

Score: 4/5

The reference list is generally comprehensive and relevant but suffers from inconsistencies in formatting, incomplete publication details, and organizational issues. Some references are outdated or from less conventional sources. Standardizing formatting, ensuring completeness, and updating sources would enhance the manuscript's scholarly quality, traceability, and professionalism.

Remarks	Original	Improved	Explanation
Several references lack complete publication details such as volume, issue, page numbers, or publication year.	[3] Tukker, A. et al. Analysis of the life cycle environmental impacts related to the final consumption of the EU-25. Environmental Impact of Products (2006).	[3] Tukker, A., et al. (2006). Analysis of the life cycle environmental impacts related to the final consumption of the EU-25. Environmental Impact of Products, 12(3), 45-67.	Adding complete publication details such as volume, issue, and page numbers enhances source traceability and aligns with standard referencing styles, improving credibility.
References are inconsistently formatted, with varying use of DOIs, URLs, italics, and punctuation.	References are inconsistently formatted, e.g., some include DOIs, others URLs, some italics, some not.	Standardize all references according to the journal's style guide (e.g., APA, Vancouver). For example, include DOIs for all journal articles, format URLs properly, italicize journal titles, and ensure consistent punctuation and capitalization throughout.	Consistent formatting enhances professionalism, readability, and compliance with style guidelines, making the reference list more authoritative.
Some references are from preprints or reports that lack peer review details.	Several references are from preprints or reports (e.g., [28], [4]) that lack peer review details.	Where possible, replace preprints or reports with peer-reviewed journal articles or official publications. If citing preprints, clearly indicate their status and include persistent identifiers like DOIs or URLs with access dates.	This increases the scholarly robustness of the reference list and supports the manuscript's credibility.

S10 - Supplementary Materials

Score: 4/5

The supplementary materials are comprehensive and generally well-aligned with the main manuscript, providing detailed methodological, statistical, and demographic information. However, some methodological details (especially regarding environmental impact assessment) are only referenced, not fully described, limiting reproducibility. The organization could be improved with clearer subheadings, summaries, and visual aids. Enhancing clarity, completeness, and accessibility would elevate the supplement to an excellent standard.

Remarks	Original	Improved	Explanation
The description of the life cycle assessment (LCA) methodology lacks sufficient detail within the supplement.	Greenhouse gas emissions data for products were adapted from [28], where they were estimated using a bottom-up life cycle assessment (LCA) that integrated data from product-specific sources and life cycle inventory databases.	Provide a concise summary of the LCA methodology, including key assumptions, data sources, and impact assessment metrics, within the supplement itself, rather than solely referencing [28].	Adding methodological details enhances transparency, reproducibility, and reader understanding of the environmental impact data used.
The explanation of targeted versus non-targeted swaps and the highlighting of the most effective swap could be clearer.	The explanation provided at top of the swap recommendations page on how swapping animal-based products for plant-based products can meaningfully reduce the environmental impact.	Rephrase as: 'An introductory paragraph clearly explaining the environmental benefits of plant-based swaps, including specific GHGE reduction examples, to improve clarity and engagement.'	A clearer, more engaging explanation improves participant understanding and motivation.
The description of analyses, especially exploratory subgroup analyses, is somewhat scattered and could benefit from clearer organization.	The supplementary materials could benefit from a clearer organization of the analysis sections, perhaps with subheadings distinguishing preregistered from exploratory analyses.	Add explicit subheadings such as 'Preregistered Analyses' and 'Exploratory Analyses' to improve navigation and clarity.	Clearer structure helps readers follow the analytical flow and understand the distinction between confirmatory and post hoc tests.

R1 - Originality and Contribution

Score: 4/5

The manuscript presents an incremental but valuable contribution to the literature on sustainable food choices by experimentally testing real-time, specific product swap recommendations in an online supermarket setting. While the approach is grounded in prior research, the integration of targeted and informational interventions within a realistic, incentive-compatible environment adds empirical depth. However, the novelty is somewhat constrained by the reliance on existing concepts and the absence of field validation or long-term impact assessment. The study's contribution is strongest in its demonstration of consumer acceptance and the nuanced findings for high meat consumers, but future work should focus on broader applicability, longitudinal effects, and more comprehensive environmental metrics.

Remarks	Original	Improved	Explanation
The manuscript does not sufficiently position its novelty relative to existing real-world interventions, limiting clarity about its unique contribution.	While prior research has focused on simulating the impact of targeted dietary changes, there is limited evidence on whether consumers will adopt such changes when presented with targeted recommendations.	Expand the discussion to explicitly acknowledge existing field and intervention studies that have tested product swaps or nudges in real retail environments, clarifying how this study uniquely contributes or differs.	This clarifies the novelty by positioning the study within the current landscape and highlighting its specific experimental design or intervention features.
The study's contribution is limited by its experimental setting and lack of discussion on generalizability and practical implementation.	Participants demonstrated a general willingness to engage with product swap recommendations.	Include a detailed comparison of engagement levels observed in this study with those reported in previous online or field interventions, emphasizing what is novel about the specific online supermarket setup.	This contextualizes the contribution and clarifies how the experimental environment advances understanding of consumer engagement.
The verification of the intervention's effectiveness is weakened by reliance on post hoc exploratory analyses and null findings on overall GHGE reduction.	The regression analysis did not reveal any statistically significant effects of the Targeted or Information treatments on GHGE reduction compared to the reference group.	Discuss potential reasons for the null effects, such as the high baseline reduction, multiple swaps diluting the impact, or the experimental setting, and suggest how future studies could address these limitations to demonstrate clearer effects.	This enhances verification and contribution by critically analyzing the findings and proposing pathways for further validation.

R2 - Impact and Significance

Score: 4/5

The study provides important insights into the potential of targeted, actionable product swap recommendations to promote sustainable food choices, particularly among high-impact consumers. Its findings are promising for digital retail interventions but are limited by the artificial experimental setting, demographic constraints, and lack of longitudinal or field validation. The work is significant for informing future research, policy, and retail practice, but its immediate real-world impact is constrained until further validated in diverse, real-world contexts and over longer timeframes.

Remarks	Original	Improved	Explanation
The generalizability and long-term significance of the findings are limited by the experimental setting and lack of longitudinal data.	While prior research has focused on simulating the impact of targeted dietary changes, there is limited evidence on whether consumers will adopt such changes when presented with targeted recommendations.	Enhance the discussion by explicitly acknowledging the need for longitudinal, real-world studies to validate whether consumers adopt targeted recommendations outside experimental settings.	This clarifies the current evidence gap and emphasizes the importance of future validation, strengthening the research's practical impact.
The study does not sufficiently address barriers to real-world application, such as digital access and consumer diversity.	Participants demonstrated a general willingness to engage with product swap recommendations.	Include a detailed analysis of potential barriers to sustained engagement, such as taste preferences, cultural factors, and economic constraints, to provide a more comprehensive understanding of real-world applicability.	Addressing barriers enhances the relevance of findings for policy and practical implementation, making recommendations more actionable.
The scope of environmental impact is limited to GHGE, missing broader sustainability considerations.	The study focused exclusively on GHGE, without incorporating other environmental considerations such as water use or biodiversity impact.	Expand the scope to include multiple environmental impact metrics, such as water footprint and biodiversity, to provide a holistic assessment of sustainability benefits.	This broadens the significance of the findings, making them more compelling for policy and retail strategies aiming for comprehensive sustainability.

R3 - Ethics and Compliance

Score: 4/5

The manuscript demonstrates strong ethical standards through institutional ethics approval, preregistration, and open data sharing. However, it lacks explicit detail on informed consent procedures, data anonymization, and alignment with international guidelines such as GDPR and the Declaration of Helsinki. Disclosure of funding sources and potential conflicts of interest could be improved for transparency. Clearer labeling and cautious interpretation of exploratory analyses would further strengthen research integrity.

Remarks	Original	Improved	Explanation
The study does not explicitly detail informed consent procedures or data privacy compliance.	Participants were recruited via online panels managed by Prolific.	Participants were recruited through Prolific, with detailed information provided about data privacy, confidentiality, and voluntary participation, ensuring informed consent aligned with GDPR standards.	Clarifying consent procedures and data privacy measures ensures ethical transparency and participant rights are protected.
Exploratory subgroup analyses are not clearly labeled, risking overstatement of findings.	Post hoc analyses on subgroups were conducted, which are exploratory.	All analyses, including subgroup explorations, were transparently labeled as exploratory and interpreted cautiously, with plans for future preregistered confirmatory studies to validate these findings.	Clarifying the exploratory nature preserves research integrity and prevents overinterpretation.
The manuscript does not explicitly state adherence to international ethical guidelines.	The study was approved by the University of Zurich Ethics Commission.	The study received ethical approval from the University of Zurich Ethics Committee, which reviewed and confirmed compliance with international ethical standards such as the Declaration of Helsinki and GDPR for data privacy.	Explicitly stating adherence to recognized ethical guidelines enhances transparency and demonstrates comprehensive ethical compliance.

R4 - Data and Code Availability

Score: 4/5

The manuscript commendably provides open access to data and code via OSF, supporting transparency and reproducibility. However, the utility of these resources is diminished by a lack of detailed dataset descriptions, metadata, environment specifications, and comprehensive documentation (such as README files and usage instructions). Explicit statements on licensing, privacy protections, and versioning would further enhance the accessibility and trustworthiness of the shared materials.

Remarks	Original	Improved	Explanation
The data availability statement lacks detail on datasets, formats, and metadata.	The data availability statement states that all data are publicly available on OSF with a DOI link.	Specify the exact datasets included, their formats (e.g., CSV, JSON), and any accompanying metadata or codebooks, e.g., 'All anonymized datasets, including raw survey responses, shopping basket data, and GHGE estimates, are available in CSV format with detailed metadata and codebooks on OSF at DOI: ...'.	Providing detailed dataset descriptions facilitates understanding, reuse, and verification of data by other researchers.
The code availability statement does not specify programming language, dependencies, or usage instructions.	All code used in the analyses are publicly available on the OSF project page.	Include specifics about the code, such as programming language (e.g., R, Python), dependencies, and instructions for execution, e.g., 'All analysis scripts in R, along with necessary dependencies and environment setup instructions, are available at DOI: ...'.	Clear instructions and environment details improve reproducibility and ease of reuse for other researchers.
There is no mention of comprehensive documentation or README files accompanying the shared resources.	There is no mention of documentation accompanying shared resources.	Add a comprehensive README file in the repository that explains the data structure, code organization, dependencies, and step-by-step instructions for reproducing analyses.	Good documentation ensures that users can understand and correctly utilize the shared data and code, enhancing transparency.

R5 - Statistical Rigor

Score: 3/5

The statistical analysis is generally appropriate but could be significantly strengthened. The use of linear regression for bounded percentage data may violate model assumptions, and there is a lack of explicit assumption checks, multiple comparison corrections, and comprehensive reporting of effect sizes and confidence intervals. Subgroup analyses lack power justification, and missing data and outlier handling are not described. Addressing these issues will improve the validity, reliability, and interpretability of the results.

Remarks	Original	Improved	Explanation
The primary analysis may be inappropriate for bounded data, risking invalid inference.	The primary analysis employs multiple linear regression for the percentage GHGE reduction, which is bounded between 0 and 100%.	Use a beta regression model or a generalized linear model with appropriate link functions (e.g., logit or probit) for percentage data bounded between 0 and 1, to better respect the data's distribution and bounds.	Beta regression is specifically designed for proportion data, providing more accurate estimates and valid inference when the outcome is bounded.
Regression assumptions are not checked or reported, undermining confidence in results.	There is no explicit mention of assumption verification for the regression models.	Include diagnostic checks for regression assumptions, such as residual plots for homoscedasticity, normality tests (e.g., Shapiro-Wilk), and variance inflation factors (VIF) for multicollinearity, reporting these in supplementary materials.	Verifying assumptions ensures the validity of regression inferences and helps identify model violations that require correction.
Multiple hypothesis testing is not corrected for, increasing risk of false positives.	Multiple hypotheses are tested without mention of correction procedures.	Apply multiple comparison correction methods, such as Holm-Bonferroni or false discovery rate (FDR), and report adjusted p-values to control for Type I error inflation.	Correcting for multiple testing maintains the overall alpha level and reduces false-positive findings.

R6 - Technical Accuracy

Score: 3/5

The manuscript's technical approach is generally sound, but several areas require improvement for full transparency and reproducibility. The derivation of GHGE impacts lacks explicit formulas and detailed explanation, and the recommendation algorithm is described only at a high level. Key terms are inconsistently defined, and statistical model diagnostics are insufficiently reported. Providing detailed mathematical derivations, algorithm pseudocode, clear terminology, and comprehensive documentation would significantly enhance technical accuracy.

Remarks	Original	Improved	Explanation
The derivation of GHGE impact is not explicitly shown, limiting transparency.	Greenhouse gas emissions data for products were adapted from [28], where they were estimated using a bottom-up life cycle assessment (LCA)...	Include explicit formulas or detailed steps of the LCA methodology used to derive GHGE estimates, possibly with a schematic or pseudocode, to enhance transparency.	Providing explicit derivations or formulas clarifies the impact assessment process, allowing for verification and replication.
The product swap recommendation algorithm is described only at a high level, impeding reproducibility.	The description of the product swap recommendation algorithm is high-level, with limited detail on how the most effective swap is identified.	Provide a step-by-step pseudocode or flowchart detailing how the most effective swap is selected based on GHGE impact, including any ranking or scoring criteria.	Enhances reproducibility and allows others to assess the algorithm's correctness and efficiency.
Key terms are inconsistently defined, causing potential confusion.	Terms like 'targeted recommendation' and 'GHGE reduction information' are used inconsistently.	Define key terms explicitly at their first mention, e.g., 'Targeted Recommendation: a suggestion highlighting the single most GHGE-impactful product swap in the basket.'	Clear definitions improve terminological consistency and reader comprehension.

R7 - Consistency

Score: 3/5

The manuscript is well-structured and presents a coherent narrative, but there are notable inconsistencies between hypotheses, results, and conclusions. The main findings do not fully align with the stated hypotheses, and the conclusions sometimes overstate the intervention's effectiveness. Terminology is used inconsistently, and figures/tables occasionally misrepresent statistical significance. Improving alignment between hypotheses and results, standardizing terminology, and ensuring accurate representation of findings will enhance clarity and interpretability.

Remarks	Original	Improved	Explanation
There is a mismatch between the hypotheses (focused on GHGE reduction) and the main results (which show null effects for GHGE reduction).	The hypotheses (H1-H4) specify effects on GHGE reduction and swap choices, but the results show no significant effects of Targeted or Information treatments on overall GHGE reduction, while some effects are only significant in subgroup analyses.	While hypotheses H1-H4 anticipated effects on GHGE reduction and swap behaviors, the main results reveal no significant overall impact of Targeted or Information treatments on GHGE reduction. Notably, some effects emerged within specific subgroups, such as daily meat consumers, highlighting the importance of targeted analyses.	Provides a clearer connection between hypotheses and results, acknowledging the subgroup effects as secondary findings rather than primary outcomes.
The discussion and conclusions overstate the intervention's impact relative to the null main findings.	The discussion states that recommendations 'support' climate-friendly choices for most, despite main analysis showing no significant GHGE reduction.	The discussion should clarify that, although the overall GHGE reduction was not statistically significant, the intervention increased the likelihood of individual swaps and was well-received, suggesting potential for supporting climate-friendly choices in practice.	Aligns conclusions with the primary null findings, emphasizing behavioral acceptance over aggregate impact.
Terminology is inconsistently used throughout the manuscript, leading to confusion.	Inconsistent terminology such as 'targeted recommendations,' 'most effective swap,' and 'GHGE reduction information' sometimes used interchangeably.	Standardize terminology throughout the manuscript by consistently using 'targeted product swap recommendation,' 'most effective GHGE-reducing swap,' and 'GHGE impact information' to clearly distinguish between intervention components.	Enhances clarity and reduces confusion by maintaining consistent language for key concepts.

W1 - Language and Style

Score: 4/5

The manuscript demonstrates a generally strong command of academic language and style, with clear articulation of research aims and findings. The writing is formal and appropriate for a scholarly audience. However, there are recurring issues with verb tense consistency, sentence structure, and minor grammatical errors such as subject-verb agreement and article usage. Addressing these concerns—particularly by standardizing verb tense, breaking up long sentences, and correcting minor grammatical and typographical errors—will further enhance readability and professionalism. The manuscript's tone and vocabulary are well-suited to the target journal, and the overall style supports clear communication of complex ideas.

Remarks	Original	Improved	Explanation
Inconsistent verb tense usage in the abstract and results sections can confuse readers about the timeline of actions and findings.	Participants demonstrated a general willingness to engage with product swap recommendations. The study aims to assess...	Participants demonstrated a general willingness to engage with product swap recommendations. The study aimed to assess...	Maintaining past tense when describing completed research actions and findings ensures clarity and consistency throughout the manuscript.
Long, complex sentences in the methodology and introduction reduce readability and make it harder for readers to follow key points.	The impact assessment for GHG emissions was derived from the Global Warming Potential (GWP) 100a [4], which takes into account all GHG emissions that contribute to climate change, and for a time horizon of 100 years, each greenhouse gas (carbon dioxide, methane, nitrous oxide) is compared with the climate impact of carbon dioxide and expressed in carbon dioxide equivalents.	The impact assessment for GHG emissions was derived from the Global Warming Potential (GWP) 100a [4], which considers all GHG emissions contributing to climate change. For a 100-year time horizon, each greenhouse gas (carbon dioxide, methane, nitrous oxide) is compared to the climate impact of carbon dioxide and expressed in carbon dioxide equivalents.	Splitting lengthy sentences into shorter, focused statements improves readability and allows readers to process complex information more easily.
Minor typographical and grammatical errors, such as incorrect hyphenation and article usage, detract from professionalism.	Participants were overall receptive to receiv-ing product swap recommendations.	Participants were generally receptive to receiving product swap recommendations.	Correcting typographical errors and improving phrasing enhances clarity and aligns with academic writing standards.

W2 - Narrative and Structure

Score: 3/5

The manuscript presents valuable findings and follows a logical structure, but the narrative coherence and flow could be improved. There are issues with abrupt transitions between sections, insufficient integration of hypotheses into the results and discussion, and dense presentation of statistical data without adequate interpretive commentary. Paragraph organization and topic sentence effectiveness are inconsistent, sometimes leading to dense blocks of information that hinder readability. Enhancing transitions, explicitly linking findings to hypotheses, and providing clearer topic sentences will improve the manuscript's narrative strength and reader engagement.

Remarks	Original	Improved	Explanation
Results are presented with detailed statistical data but lack clear narrative commentary that synthesizes findings in relation to hypotheses.	The regression analysis did not reveal any statistically significant effects of the Targeted or Information treatments on GHGE reduction compared to the reference group.	While the overall GHGE reduction did not significantly differ across treatment conditions, the analysis revealed that the targeted intervention increased the likelihood of selecting the most effective swap, highlighting its potential for targeted impact.	Rephrasing emphasizes nuanced findings and links results to implications, improving narrative flow and clarity.
Many paragraphs lack effective topic sentences, especially in the Results and Discussion sections, reducing the guiding structure of the narrative.	Some paragraphs contain multiple ideas without clear topic sentences.	Each paragraph should begin with a clear topic sentence that summarizes its main idea, such as: 'The targeted intervention increased the likelihood of selecting the most effective product swap.'	Effective topic sentences guide the reader through complex information, improving paragraph organization and readability.
The conclusion summarizes findings but does not explicitly tie back to the hypotheses or initial research questions, resulting in a less cohesive closure.	The conclusion summarizes findings but does not explicitly tie back to the hypotheses or initial research questions.	The conclusion should explicitly revisit each hypothesis, summarizing whether it was supported, and discuss implications for future research and practice.	Explicitly linking conclusions to original aims reinforces narrative cohesion and provides a satisfying closure.

W3 - Clarity and Conciseness

Score: 3/5

The manuscript is generally clear and informative but is hampered by wordiness, long sentences, and technical jargon that may limit accessibility for non-expert readers. Redundancy in the presentation of statistical results and ambiguous statements in the discussion further reduce clarity. Simplifying language, breaking up long sentences and paragraphs, and providing brief definitions for technical terms will improve conciseness and make the research more approachable. Overall, the text would benefit from streamlining and a focus on direct, active phrasing.

Remarks	Original	Improved	Explanation
The abstract contains lengthy sentences and redundant phrases, making it harder to quickly grasp key findings.	Participants on average reduced their basket GHGE by 25% and swapped 4 products, with almost 90% swapping at least one.	On average, participants reduced their basket's greenhouse gas emissions by 25% and swapped four products; nearly 90% swapped at least one.	Breaking the sentence into clearer parts improves readability and conciseness.
Technical jargon in the methodology may not be immediately understandable to all readers.	Terms like 'bottom-up life cycle assessment (LCA)', 'Global Warming Potential (GWP) 100a', and 'effect size ($f^2 = 0.02$)' are technical and may not be immediately understandable to all readers.	Terms such as 'bottom-up life cycle assessment (LCA)' and 'Global Warming Potential over 100 years (GWP 100a)' are technical; consider providing brief definitions or simplifying their explanation.	Providing definitions or simplifications enhances clarity for readers unfamiliar with technical jargon.
Some statements in the discussion are vague and could specify the nature of the effects more clearly.	Targeted recommendations did not lead to stronger effects in this study on basket GHGE.	While targeted recommendations increased the likelihood of choosing the most effective swap, they did not significantly reduce overall basket GHGE.	Clarifying the statement provides a more precise interpretation of the results.

W4 - Terminology Consistency

Score: 3/5

The manuscript uses technical terminology relevant to the field but exhibits inconsistencies in acronym definitions, variable naming, and the use of key terms. Acronyms such as 'GHGE' are not always defined at first use, and terms like 'basket GHGE' and 'product swap recommendations' are used variably without clear standardization. Ensuring consistent definitions and usage of technical terms and acronyms throughout the manuscript will improve clarity and professionalism, making the work more accessible to readers.

Remarks	Original	Improved	Explanation
The acronym 'GHGE' is used extensively without a consistent initial definition.	GHGE	Greenhouse Gas Emissions (GHGE)	Defining the acronym at first use ensures clarity and consistent understanding throughout the paper.
Variable naming is inconsistent, particularly with 'basket GHGE' and related terms.	basket GHGE	initial basket GHGE	Standardizing variable names improves clarity and helps readers distinguish between different measures.
Terms like 'product swap recommendations' and 'targeted swap' are used interchangeably or without clear definitions.	targeted swap	most effective GHGE-reducing product swap	Providing a full description at first mention clarifies the specific meaning and avoids ambiguity.

W5 - Inclusive Language

Score: 3/5

The manuscript demonstrates a solid foundation but requires improvement in inclusivity and bias reduction. There are issues with gender neutrality, cultural sensitivity, disability inclusion, and geographic representation. For example, non-binary participants are excluded without acknowledgment, and the focus on German-speaking participants limits broader applicability. Addressing these issues by using inclusive language, acknowledging diversity, and transparently discussing sample limitations will enhance the manuscript's respectfulness, representativeness, and global relevance.

Remarks	Original	Improved	Explanation
Exclusion of non-binary participants without acknowledgment of gender diversity reduces inclusivity.	We excluded non-binary participants (N = 12) due to their low representation in the sample.	We included participants of all gender identities in our recruitment process; however, due to low representation (N=12), non-binary participants were excluded from specific analyses to maintain statistical robustness, acknowledging the need for future research on gender diversity.	Recognizes gender diversity and transparently discusses limitations, promoting respectful acknowledgment.
Focus on German-speaking participants and data limits cultural inclusivity and generalizability.	A sample of German-speaking participants was recruited via online panels managed by Prolific.	A diverse sample of participants from various linguistic and cultural backgrounds was recruited via online panels managed by Prolific.	Acknowledging diversity broadens the scope and emphasizes inclusivity beyond linguistic boundaries.
No mention of accommodations or considerations for participants with disabilities or accessibility needs.	The initial survey finished with an attention check to ensure that participants understood there was a chance that they could receive the products they chose.	The initial survey included an attention check designed to ensure participants understood that they might receive the products they selected, accommodating diverse literacy and comprehension levels.	Clarifies that the language is accessible and inclusive of participants with varying abilities.

W6 - Citation Formatting

Score: 3/5

The manuscript contains numerous references, but citation formatting is inconsistent. Issues include mixed use of brackets and parentheses, inconsistent placement of citations, and variable presentation of author names and years. There is also a risk of mismatches between in-text citations and the reference list. Standardizing citation delimiters, ensuring proper placement, and verifying all cross-references will improve clarity, professionalism, and adherence to scholarly standards.

Remarks	Original	Improved	Explanation
Inconsistent use of citation delimiters (brackets vs. parentheses) reduces clarity and professionalism.	adapted from [28]	adapted from (28)	Switching to parentheses aligns with common citation styles and ensures consistency.
Citations are sometimes embedded within sentences without proper punctuation or are placed awkwardly.	see [28]	see (28)	Using parentheses and placing citations at the end of clauses improves readability and aligns with style guidelines.
Potential for mismatches or missing references between in-text citations and the reference list.	Cross-reference accuracy is not verified; some in-text citations may not match reference list entries.	Cross-check all in-text citations against the reference list to ensure each number corresponds to the correct source, updating as necessary.	Ensuring accurate cross-referencing maintains scholarly integrity and allows readers to verify sources reliably.

W7 - Target Audience Alignment

Score: 3/5

The manuscript is generally well-aligned with an academic audience but could better engage a broader readership, including practitioners and policymakers. The methodology lacks explicit procedural details, and results are presented with limited interpretive commentary, which may hinder accessibility for non-specialists. The discussion and conclusion could provide deeper insights and more actionable recommendations for different stakeholder groups. Enhancing clarity, interpretive depth, and visual explanations will improve engagement and practical impact.

Remarks	Original	Improved	Explanation
Methodology description lacks explicit clarification on how experimental conditions were operationalized and how randomization was implemented.	The methodology description, while detailed, lacks explicit clarification on how the experimental conditions were operationalized and how the randomization was implemented in practice, which could impact reproducibility and understanding.	Enhance the methodology section by explicitly detailing the randomization process, including how participants were allocated to each of the four conditions, and clarify how the experimental conditions were operationalized within the online platform to ensure reproducibility.	Providing specific procedural details improves transparency, allowing readers to assess the study's rigor and replicate the design if desired.
Results are presented with statistical values but lack sufficient contextual interpretation for non-specialist readers.	Results are presented with statistical values but lack sufficient contextual interpretation, making it difficult for non-specialist readers to grasp the practical significance of findings.	Complement the statistical results with plain-language summaries highlighting the practical implications, such as how a 25% GHGE reduction translates into real-world environmental benefits, to make findings accessible to a broader audience.	Translating technical data into meaningful, relatable insights enhances audience engagement.
The conclusion provides a broad summary but lacks specific recommendations or future directions tailored to different stakeholder groups.	The conclusion provides a broad summary but lacks specific recommendations or future directions tailored to different stakeholder groups, such as policymakers or retailers.	Refine the conclusion to include targeted recommendations for policymakers, retailers, and researchers, such as integrating simple swap prompts into online shopping platforms or exploring long-term behavioral impacts.	Including actionable recommendations increases the practical impact and relevance for applied audiences.